

Current Trends in the Management of Hiatal Hernia: A Literature Review of 10 Years of Data.

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Hiatal hernia (HH) is commonly detected during endoscopic examinations and is associated with gastroesophageal reflux disease. In recent years, there have been significant advancements in diagnosing and treating HH. Surgical techniques for HH repair include open surgery, various laparoscopic procedures, transoral incisionless fundoplication, and magnetic sphincter augmentation (MSA). Laparoscopic Nissen fundoplication is often considered the standard for treating gastroesophageal reflux disease-related HH due to its effectiveness. Other procedures, such as Toupet and Dor funduplications, may be suited for patients with specific conditions, such as impaired esophageal motility. Newer approaches, including the MSA system and mesh repair, focus on patient-specific treatments to achieve the best outcomes. This review synthesizes the literature from 2014 to 2024 to provide an overview of current trends in HH management.

Laparoscopic Nissen repair: indications, techniques and long-term benefits.

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The Nissen fundoplication or total 360 degrees fundoplication is probably the most frequently used anti-reflux procedure throughout the world. With the advent of laparoscopic surgery the popularity among surgeons to perform a laparoscopic Nissen fundoplication has even increased. The purpose of this paper is to provide an overview of the experience of laparoscopic Nissen fundoplication over the past 15 years. We performed an extensive review of the literature in order to ascertain the representative papers. In addition, available consensus papers, especially with regard to indication and technique, were assessed. Indication for a laparoscopic Nissen fundoplication should depend on documentation of the presence of disease as well as objective testing of the functional disorders and the complications. The technique of Nissen fundoplication is discussed controversially. Consensus exists with regard to floppiness of the wrap, necessary closure of the crurae and the use of a calibration method during the performance of the wrap. The laparoscopic technique creates a learning curve, which needs to be respected. Large prospective series in recent years have shown a complication rate between 5% and 10%, depending on the definition of the complication. In these last prospective series good and excellent results have been reported, of between 85% and 95%. Reflux recurrence is reported as between 1% and 8.5%, with a concomitant dysphagia rate of 0%-10%. The Nissen fundoplication is currently performed throughout the world, most frequently in a minimally invasive technique. Several randomized trials that have been performed in the past years document that the Nissen fundoplication is an effective procedure for the treatment of pathological gastro-oesophageal reflux disease when a critical indication is used for well-defined patients.

Watson fundoplication in children: a comparative study with Nissen fundoplication.

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Nissen fundoplication is the gold standard antireflux procedure in children. In 1996, one pediatric surgeon adopted the anterior fundoplication described by Watson in 1991. This procedure is reported to achieve

good reflux control while permitting burping, active vomiting, and reducing gas bloat. An audit project was undertaken to compare the clinical outcome of children undergoing Nissen and Watson fundoplication. The case notes of 144 children undergoing open fundoplication between February 1995 and February 2002 were reviewed retrospectively. Results of 72 boys and 59 girls comprising 76 Nissen and 55 Watson fundoplications were assessed. In each group, one death occurred within 1 month of operation. Chest infections occurred in 6.6% (Nissen) and 1.8% (Watson), and wound infections in 2.6% and 1.8%, respectively. Dysphagia was recorded in 7.9% of Nissen and 1.8% of Watson fundoplications. Follow-up data were analyzed in 70 children with Nissen and 48 children with Watson fundoplication. When overall clinical outcome was assessed for those patients with a minimum follow-up of 1 year, 85.1% Nissen and 88.2% Watson were judged good/excellent; 14.9% Nissen and 11.8% Watson were judged poor/bad. Watson fundoplication can safely be performed in children with comparable clinical outcome to Nissen fundoplication.

A novel antireflux procedure: gastroplasty with restricted antrum to control emesis (GRACE).

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Nissen fundoplication is the most commonly performed operation to treat gastroesophageal reflux disease and vomiting in children with neurologic impairment. However, failure rates of Nissen fundoplication in this population are higher, and alternatives to Nissen fundoplication have technical and functional disadvantages. We hypothesize that the novel gastroplasty with restricted antrum to control emesis (GRACE) would be more effective than Nissen fundoplication at reducing emetic reflux. To compare the GRACE with Nissen fundoplication, 15 canine subjects were randomized to Nissen fundoplication or GRACE. All subjects underwent gastrostomy tube placement. Baseline gastric emptying, electrogastrography, and induced vomiting studies were performed. Nissen fundoplication or GRACE was then performed. Postoperatively, gastric emptying and vomiting studies were repeated. Gastric emptying before and after antireflux procedures was not significantly different between groups. Both Nissen fundoplication (38%, $P = .04$) and GRACE (69%, $P < .01$) procedures prevented reflux compared with baseline. However, the GRACE procedure significantly reduced reflux when compared with Nissen fundoplication ($P = .03$). In this canine model, GRACE appears to be significantly more effective than Nissen fundoplication at reducing emetic reflux. This novel procedure preserves gastric function and is well tolerated. The GRACE procedure may provide an alternative to Nissen fundoplication as a primary or repeat antireflux procedure for children with neurologic impairment.

Partial versus complete fundoplication: is there a correct answer?

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Gastroesophageal reflux disease is a common disorder, and patients diagnosed with GERD face a lifelong treatment requirement. A surgical antireflux procedure may be offered as an alternative to lifelong treatment with proton-pump inhibitors. Many investigations have been performed to help discover the best surgical alternative to medical management. An ideal antireflux procedure should be safe, effective, durable, and result in minimal complications. Total fundoplication in the form of Nissen fundoplication is the most widely used antireflux operation worldwide. Although its efficacy is well documented, the

clinical success rate in terms of reflux control is occasionally compromised by troublesome mechanical side effects. Because of these unsatisfactory symptoms and continued hindered quality of life, the Nissen fundoplication has undergone many modifications. The current standard appears to be the 2 cm floppy Nissen; however, the alternative approach has been the use of a partial fundoplication, most frequently the Toupet procedure. Both the Nissen and Toupet fundoplications have proven to provide relief in the majority of patients, but each has its own drawback. Patients undergoing Nissen fundoplication have a higher incidence of dysphagia early after operation, although this appears to resolve in most. The Toupet, on the other hand, may not be as durable, and may lead to the early re-emergence of symptoms. The problem of post-Nissen dysphagia led many surgeons to believe that the Nissen might be contraindicated in patients who have dysmotility, because it would cause even greater dysphagia; however, recent articles have not demonstrated this to be the case. It seems that the floppy Nissen performed over a large bougie (56-60 Fr) with division of short gastrics and crural closure is an acceptable operation for reflux in both those who have normal motility and those who have mild to moderate dysmotility. Thus, for most patients who have GERD and normal motility, either procedure appears effective in the majority of patients; however, those patients who have severe dysmotility disorders and who require an antireflux procedure (ie, scleroderma, postmyotomy achalasia) are likely best served with a partial fundoplication.

Interventional anti-reflux management for gastro-oesophageal reflux disease in lung transplant recipients: a systematic review and meta-analysis.

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Gastroesophageal reflux disease (GORD) and aspiration are risk factors in the development of bronchiolitis obliterans syndrome (BOS) in the lung transplant population. The aim of this study was to investigate if allograft function and survival improved after anti-reflux surgery (ARS) in lung transplant recipients. In accordance with PRISMA guidelines, we conducted a systematic search of MEDLINE, Embase, and the Cochrane library databases from inception until 13/01/2024. Articles reporting outcomes of ARS following lung transplantation were included. A random effects model was used for meta-analysis. The search identified 20 which were used for quantitative analysis. Overall, FEV1 and rate of change of FEV1 had improved following ARS by 0.141 L/s (95% CI; -0.282, -0.001) and -1.153 mL/d (95% CI; -12.117, -0.188), respectively. Survival hazard ratio post-ARS was 0.39 (95% CI; 0.19, 0.60). Nissen fundoplication was the most effective anti-reflux procedure with the greatest effect on reduction in the rate of change of FEV1, with an improvement of -2.353 mL/d (95% CI; -3.058, -1.649). ARS in lung transplant recipients improves allograft function and survival. Given the increased incidence of GORD in lung transplant recipients, there should be a low threshold for investigation of GORD and subsequent ARS.

Robotic versus laparoscopic revision to Toupet fundoplication for failed Nissen fundoplication: a single-center experience.

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Nissen fundoplication (NF) is a common surgical procedure to treat gastroesophageal reflux disease; however, a subset of patients may continue to experience symptoms or develop symptom recurrence despite a successful procedure. This study aims to compare laparoscopic and robotic approaches for

treating failed NF and evaluate the outcomes after converting to Toupet fundoplication (TF). We conducted a retrospective analysis of patients who underwent robotic or laparoscopic revision to TF for failed NF between 2016 and 2023. The data collected included demographics, pre-operative workup, and peri- and post-operative outcomes. Symptom analysis and anti-reflux medication usage were collected using a patient questionnaire. Failed fundoplication was defined as the need for an additional operation due to unresolved GERD symptoms or the emergence of a new issue. Eighty-eight patients (56 laparoscopic, 32 robotic) were included. Mean operative time was 148.71 ± 53.64 min for the total cohort and was significantly longer in the robotic group (RG) 167.43 min vs 138.01 min in the Laparoscopic group (LG) (p value = 0.012). The LG had a length of hospital stay of 2.16 ± 1.69 days vs RG 2.21 ± 1.28 days (p value = 0.867). The LG had a higher number of early readmissions (5.4%, p value = 0.629) and both the LG and the RG had 1 patient that required an early reintervention. Symptoms of dysphagia and reflux decreased in both groups at last follow-up, but the reduction in PPI use was not significant. Surgical revision to TF for failed NF provides significant symptom improvement with low rates of complications and recurrences. Our study shows that both approaches are safe and feasible and have comparable surgical and symptom outcomes.

Prosthetic patch stabilization of crural repair in antireflux surgery in children.

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Nissen fundoplication is a frequently performed procedure by pediatric surgeons for the treatment of gastroesophageal reflux. Reoperation for failed fundoplication is necessary in 10 per cent of children and in 25 per cent of neurologically impaired children. In an attempt to reduce the postoperative recurrence of gastroesophageal reflux and wrap herniation, we have modified our Nissen fundoplication by reinforcing the crural repair with a horseshoe-shaped prosthetic patch. Between 1993 and 1995, 38 children underwent a Nissen fundoplication with prosthetic patch crural repair. Tension-free crural repair was performed over an esophageal dilator by suturing a horseshoe-shaped Dacron patch posterior to the esophagus. After crural repair, the standard Nissen 360 degrees wrap was constructed. Of the 38 children, 56 per cent ($n = 21$) were neurologically impaired. In 18 per cent of patients, the operation was performed for recurrent reflux due to wrap herniation after a Nissen fundoplication without prosthetic patch repair. One child required esophageal dilatation postoperatively for dysphagia related to esophageal stenosis. After a mean follow-up of 15 months, all patients were without symptomatic reflux or had no radiographic evidence of recurrent reflux or wrap herniation. Tension-free crural repair with a prosthetic patch may decrease postoperative reflux and wrap herniation, particularly in neurologically impaired children and in children requiring reoperation for recurrent reflux.

Acquired volvulus following Nissen fundoplication.

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Nissen fundoplication is the procedure of choice for the management of gastroesophageal reflux disease. We report a case of acquired gastric volvulus following open fundoplication. The mechanism of formation and correction of the volvulus is discussed.

Nissen vs Toupet laparoscopic fundoplication.

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Nissen fundoplication (360 degrees) is the standard operation for the surgical management of gastroesophageal reflux disease (GERD). To avoid postoperative dysphagia, it has been proposed that antireflux surgery be tailored according to the degree of preexisting esophageal motility. Postoperative dysphagia is thought to occur more commonly in patients with esophageal dysmotility and the Toupet procedure (270 degrees) has been recommended for these patients. We performed a randomized trial to evaluate this tailored concept and to compare the two operative techniques in terms of reflux control and complication rate (dysphagia). Our objective was to determine the impact of preoperative esophageal motility on the clinical and objective outcome, following Toupet vs Nissen fundoplication and to evaluate the success rate of these procedures. From May 1999 until May 2000, 200 patients with GERD were included in a prospective randomized study. After preoperative examinations (clinical interview, endoscopy, 24-h pH study and esophageal manometry), 100 patients underwent either a laparoscopic Nissen (50 with and 50 without motility disorders), or a Toupet procedure (50 with and 50 without motility disorders). Postoperative follow-up after 4 months included clinical interview, endoscopy, 24-h pH study and esophageal manometry. Interviews showed that 88% (Nissen) and 90% (Toupet) of the patients, respectively, were satisfied with the operative result. Dysphagia was more frequent following a Nissen fundoplication than after a Toupet (30 vs 11, $p < 0.001$) and did not correlate with preoperative motility. In terms of reflux control, the Toupet proved to be as effective as the Nissen procedure. Tailoring antireflux surgery to esophageal motility is not indicated, since motility disorders are not correlated with postoperative dysphagia. The Toupet procedure is the better operation because it has a lower rate of dysphagia and is as effective as the Nissen fundoplication in controlling reflux.